

Segway

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R3435

2025

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- Segway —

Segway

Segway

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- $M = 10$
- $m = 2$
- $L = 0.5$
- $R = 0.1$

$$\mathbf{x} = [\theta, \dot{\theta}, \phi, \dot{\phi}, x, \dot{x}]^T$$

$$\mathbf{M}(\theta) \begin{bmatrix} \ddot{\theta} \\ \ddot{\phi} \end{bmatrix} + \mathbf{C}(\theta, \dot{\phi}) + \mathbf{G}(\theta) = \begin{bmatrix} u \\ -u \end{bmatrix}$$

$$: n = 2$$

$$: m = 1$$

$$: m < n$$

$$\lim_{t \rightarrow \infty} \theta(t) = 0, \quad \lim_{t \rightarrow \infty} \dot{\theta}(t) = 0$$

- : $|u(t)| \leq u_{\max}$
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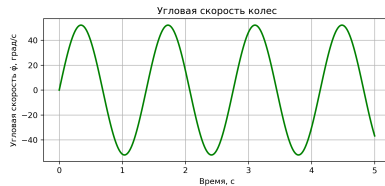
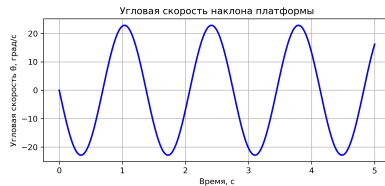
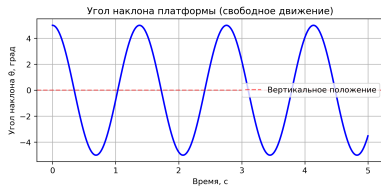
$$\mathbf{y} = [\theta, \dot{\theta}]^T$$

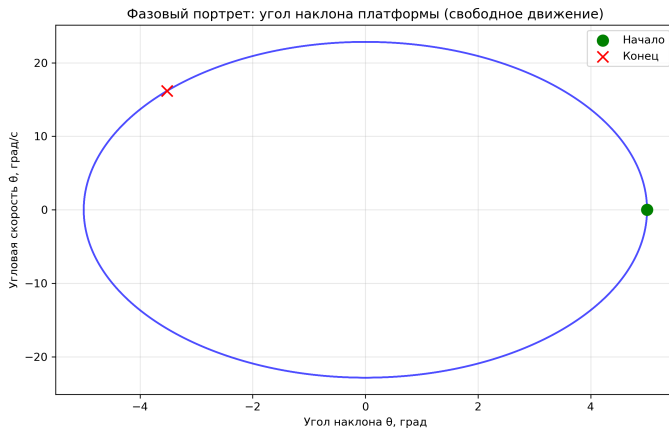
IMU (+)

$$u = -K_p\theta - K_d\dot{\theta}$$

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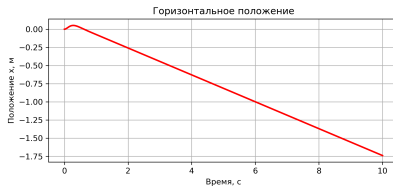
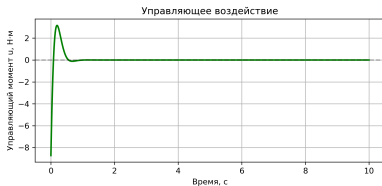
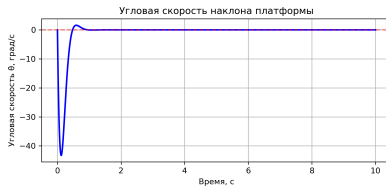
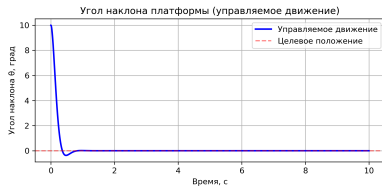


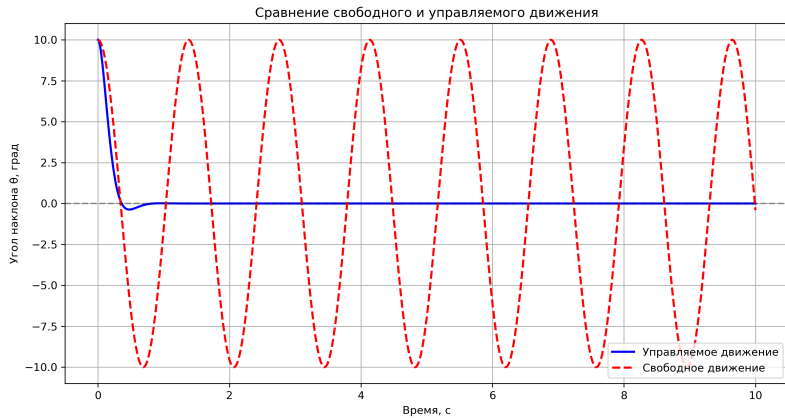


$$\mathbf{M}(\theta) \begin{bmatrix} \ddot{\theta} \\ \ddot{\phi} \end{bmatrix} + \mathbf{C} + \mathbf{G} = \begin{bmatrix} -K_p\theta - K_d\dot{\theta} \\ K_p\theta + K_d\dot{\theta} \end{bmatrix}$$

- $K_p = 50.0$ —
- $K_d = 10.0$ —

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- : $< 0.000001^\circ$
-

1° : 0.283

: 0.16 ()

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Segway

$$: u = -K_p\theta - K_d\dot{\theta}$$

Segway





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